

RESEARCH PLAN – Economics Complexity

Name of the Researchers:	Bruinsma, Patrik; Virág, Ákos
Studies and faculty:	Social Data Science, Master of Science
Course:	Economic Complexity
Title of the Research:	Do regions with more brokers produce more atypical technology combinations?

GREAT IDEA BUT NEEDS A
REWRITE

1. Short introduction

Innovation often emerges from recombination of existing technologies, but most recombinations are highly conventional. Only a small fraction of inventions relies on atypical combinations of knowledge domains, and these tend to be associated with high-impact outcomes (Uzzi et al., 2013; Kim et al., 2016). Recent regional innovation research connects co-inventor network structure to such atypical combinations. Abbasiharofteh et al. (2023) show that European regions where bridging ties create short paths between cohesive co-inventor communities tend to generate a growing share of atypical patents. Their focus, however, is mainly on meso-level structures (communities, bridging ties, small-worldness), not on who occupies key positions in these networks.

At the same time, Tóth and Lengyel (2021) provide firm-level evidence that incoming inventor brokers, who are inventors occupying brokerage positions in co-inventor networks, significantly increase the citation impact of the hiring firm's patents, especially when the firm's internal collaboration network is small-world. Our research aims to bridge these strands by asking, whether at the regional level, do regions with more inventor brokers in their co-inventor networks generate more atypical technology combinations in subsequent years, and does this depend on the region's small-world structure?

there is more
influential
work
on
brokerage

But

1. a) Novelty

Place it more behind $\Rightarrow$ impact

The project is novel in three ways:

1. It extends the brokerage mechanism identified by Tóth and Lengyel (2021) from firm-level patent impact to regional-level atypical combinations.
2. It complements Abbasiharofteh et al. (2023) by shifting attention from community-level bridging ties to individual inventors in brokerage positions.
3. It explicitly separates broker density (who sits where) from network architecture (small-worldness) when explaining atypical combinations.

what is this?

explain more

* Brokerage and cohesive ties complement each other.
Try to reflect on this too

In this way we offer a more detailed account of how actor positions and network structure jointly shape regional technological novelty.

1. b) Motivation

From a regional development perspective, policy increasingly emphasizes “smart specialization” and diversified, but related regional knowledge bases, rather than simple R&D intensity. If atypical technology combinations are key to radical innovation, understanding which network mechanisms support them becomes crucial. The questions arise which collaborations should be fostered, which boundary-spanning inventors (“brokers”) matter most, and when “more ties” are not enough without specific structural roles.

From a course perspective (Economic Complexity & Networks / Social Data Science), the topic sits at the intersection of network structure (small-worldness, brokerage, core-periphery), regional innovation and complexity (knowledge space, relatedness, novelty), and the micro-macro link (individual positions vs regional outcomes). For us, the project is also attractive because it requires a full Social Data Science pipeline with a large-scale patent data processing, a network construction, and a panel regression with carefully defined novelty and brokerage measures.

1. c) Connection to the topic

The course focuses on how network structure and economic complexity shape regional innovation and growth. Our project connects directly to three central themes: small-world networks in innovation; brokerage and structural holes; atypical combinations and complexity.

- Schilling & Phelps. (2004) argue that small-world knowledge networks with dense clusters and short paths support more and more knowledge creation. Abbasiharofteh et al. (2023) find positive linear relationship between small-worldness and the growth of atypical patents; the quadratic term is not significant.
- Following Burt (1992, 2004), actors bridging “structural holes” access non-redundant knowledge and may be more innovative. Tóth and Lengyel (2021) confirm this for mobile inventor brokers and firm-level patent impact, especially in small-world internal networks.
- Uzzi et al. (2013) show that mixing conventional content with some atypical combinations increases scientific impact; Kim et al. (2016) find this on patents. Recent science-of-science and complexity research (e.g. Fontana et al., 2020) emphasizes the novelty and interdisciplinarity of scientific knowledge.

instead of listing literature, try to develop a line of argument

Our project embeds brokerage and small-worldness into this economic complexity narrative in which regions that combine diverse knowledge through well-positioned brokers may move towards more complex and high-value specializations.

2. Short literature review

⇒ make it more a story and less a list

We highlight three strands and the remaining gap: co-inventor networks and regional innovation; brokerage at the firm and individual level; atypical combinations and novelty.

Co-inventor networks, derived from patent data, map the socio-professional networks underlying regional knowledge production (Breschi & Lenzi, 2016; Lobo & Strumsky, 2008). Fleming et al. (2007) show that local co-inventor networks with short average path lengths are associated with higher regional patenting. Tóth et al. (2021) show that repeated inter-regional co-inventor collaborations are strongly geographically structured, and discuss their potential implications for regional innovation outcomes, underlining that both who collaborates and how often collaborates is important

Burt's (2004) structural holes theory argues that brokers can recombine diverse information. Juhász, Tóth and Lengyel (2020) find that core-periphery brokers in the Hungarian film industry have a higher probability of creative success (winning awards) if they belong to the network core and bridge the core to the periphery. Tóth and Lengyel (2021) show that firms hiring inventor brokers achieve higher patent citation impact, and this broker effect is stronger when the firm's internal co-inventor network has a small-world structure.

Uzzi et al. (2013) define atypical combinations as unusual pairings of prior knowledge domains and show that "conventional + atypical" combinations underpin high-impact science. Kim et al. (2016) extend this to patents via unusual IPC combinations. Abbasiharofteh et al. (2023) apply Uzzi-type measures of atypical combinations to regional co-inventor networks and show that regions where bridging ties connect specialized co-inventor communities have a growing share of atypical patents.

We know that bridging ties and small-world structures matter, but much less thought about the individual inventors who carry these connections at the regional level. The role of inventor brokers is well documented at the firm level (Tóth & Lengyel, 2021), but not systematically linked to regional atypical combinations. Moreover, specific brokerage types – such as core-periphery or international brokers – may matter more than generic centrality, yet this has not been fully explored in a regional patent context. Our research addresses these gaps.

- * - do not mention authors, only cite them
- group literature into paragraphs that are building blocks of your story

3) Objective, methodology

3. a) Main Research question

Do regions with more inventor brokers in their co-inventor networks produce more atypical technology combinations?

- RQ1: Is a higher share of internal brokers (inventors connecting disconnected local communities) associated with a higher share of atypical patents?

- RQ2: Is a higher share of external (international) brokers (inventors connecting the region to extra-regional knowledge) associated with a higher share of atypical patents?

- RQ3: How do these two brokerage types interact? Do they have a complementary effect?

- RQ4: Is the effect of brokers (of either type) stronger in regions with small-world co-inventor networks, as suggested by Tóth and Lengyel (2021)?

Hypotheses:

- H1: Regions with a larger share of internal brokers produce a higher share of atypical technology combinations.
- H2: Regions with a larger share of external (international) brokers produce a higher share of atypical technology combinations.
- H3: The effects of internal and external brokerage are complementary (a positive interaction). The positive effect of external brokers is strongest in regions that also have a high share of internal brokers as this internal network is needed to absorb and diffuse the new, external knowledge.
- The positive effect of broker share (for both types) on atypical combinations is stronger in regions with medium–high small-worldness.

3. b) Data.

We will use European Patent Office data (e.g. PATSTAT or OECD REGPAT) for EU27/EEA regions, roughly 1990–2015. Patents are assigned to NUTS-2 regions via inventor addresses (fractional counting if needed). Technologies are defined by IPC codes.

3. c) Networks

First, for each region and time window (e.g. 5-year rolling) we build an internal co-inventor network: nodes are inventors with at least one patent in the region, edges link inventors who co-appear on a patent. Internal networks are the baseline; robustness checks may use ego-networks including cross-regional ties.

2Q2 is about external ties

Second, for each inventor we compute Burt's constraint. We identify brokers as inventors with low constraint within their region-year (e.g. top decile). Following Juhász et al. (2020), we also identify core-periphery brokers by combining coreness measures with gatekeeping roles. Aggregated regional indicators include broker share, core-periphery broker share, and optionally international broker share (inventors collaborating across multiple countries).

Third, for each region-year network we calculate clustering, average path length (giant component), and a small-world index comparing these to random graphs (Watts & Strogatz, 1998; Fleming et al., 2007). We allow for non-linear effects of this index.

Fourth, following Uzzi et al. (2013) and Abbasiharofteh et al. (2023), we compute z-scores for IPC pair co-occurrences based on historical frequencies. For each patent we derive novelty measures (e.g. low-percentile z-score) and classify patents as atypical if they include at least one pair with negative z-score (following Abbasiharofteh et al., 2023). At the region-year level, our outcome is the share of atypical patents.

* Last but not least, we estimate panel regressions at the region-year level, with the share of atypical patents as dependent variable. Key regressors are lagged broker share, small-world index, and their interaction, with controls for patent intensity, technological diversity and basic regional characteristics. Region and year fixed effects absorb unobserved heterogeneity and common shocks. Robustness checks include alternative broker definitions, alternative novelty thresholds, and, if feasible, patent-level models.

include the estimation equation

3. d) Risk Management

We identify the main risks and summarize mitigation strategies:

- *Data access and scale.* PATSTAT is large and demanding, which we mitigate by focusing on a subset of years or technologies, using pre-processed regional patent datasets if available.

* How will this be different from Abbasiharofteh et al. ?
↳ Instead, you can look at the individual or firm level ?
↳ Brokers can be local or interregional ← !!

- *Computational complexity.* Network measures for many region–year combinations can be heavy, therefore we use efficient network libraries, focus on the giant component, and approximate betweenness if needed.
- *Novelty measures.* Uzzi-type z-scores are technically demanding, so we follow Abbasiharofteh et al. (2023) closely and, if time is tight, we begin with an alternative novelty indicator (e.g. Verhoeven et al., 2016) and refine later.
- *Time constraints.* The full pipeline may exceed course deadlines, so we prioritise a minimal version that delivers descriptive network statistics and a baseline regression for H1–H2; broker types and international brokerage are treated as extensions for later publication purposes.

4. Expected outcomes

We expect empirical and theoretical outcomes.

We anticipate a positive association between regional broker share and the share of atypical patents (H1). In line with Tóth and Lengyel (2021), we expect a broker $\times$ small-worldness interaction (H2), with brokers having stronger effects in small-world-like regional networks. Exploratorily, we expect some broker types (core–periphery and international brokers) to be more strongly associated with atypical combinations than generic brokerage (H3).

The project will extend the firm-level mechanism identified by Tóth and Lengyel (2021) to the regional level and to atypical combinations rather than citation impact. It will refine arguments about small-world networks by showing that their benefits may depend on the presence and density of brokers, and clarify which brokerage roles matter for regional radical innovation.

5. Usability of research results

The results of this research would be directly useful for regional development agencies, cluster organisations, science parks, firms, universities and national or EU-level policymakers. If we can show that regions with a higher share of well-positioned inventor brokers in small-world-like co-inventor networks systematically generate more atypical, high-value patent combinations, these actors gain concrete levers instead of generic “networking” advice. Regional agencies and smart specialisation authorities could use broker density, small-world indices and atypical patent shares as early indicators of radical innovation potential, monitor whether their regions are drifting into technological lock-in, and design funding calls that deliberately connect currently separated

technological communities through identified brokers. Cluster managers and innovation intermediaries could move from broad, undirected networking events towards targeted matchmaking that puts specific core–periphery or international brokers into central roles in consortia, advisory boards and joint R&D projects, increasing the chances that funded collaborations actually recombine distant knowledge domains.

6. Summary

Redundant

This research plan proposes to investigate whether regions with more inventor brokers in their co-inventor networks produce more atypical technology combinations in patents, and whether this depends on the small-world structure of those networks. Building on Tóth and Lengyel (2021) and Abbasiharofteh et al. (2023), we move to the regional level and focus on the positions of individual inventors within regional co-inventor networks.

Using European patent data, we will construct region–year co-inventor networks, measure brokerage and small-worldness, compute Uzzi-type novelty measures, and estimate panel models relating brokerage to atypical patent shares. If successful, the project will improve our understanding of how brokerage and network structure jointly shape regional radical innovation, provide policy-relevant insights for regional innovation strategies.

7. Important bibliography

- Abbasiharofteh, M., Kogler, D. F., & Lengyel, B. (2023). Atypical combinations of technologies in regional co-inventor networks. *Research Policy*, 52(10), 104886. <https://doi.org/10.1016/j.respol.2023.104886>
- Breschi, S., & Lenzi, C. (2016). Co-invention networks and inventive productivity in US counties. *Journal of Regional Science*, 56(1), 52–91. <https://doi.org/10.1016/j.jue.2015.12.003>
- Burt, R. S. (2004). Structural holes and good ideas. *American Journal of Sociology*, 110(2), 349–399. <https://doi.org/10.1086/421787>
- Fontana, M., Iori, M., Montobbio, F., & Sinatra, R. (2020). New and atypical combinations: An assessment of novelty and interdisciplinarity. *Research Policy*, 49(7), 104063. <https://doi.org/10.1016/j.respol.2020.104063>

- Juhász, S., Tóth, G., & Lengyel, B. (2020). Brokering the core and the periphery: Creative success and collaboration networks in the film industry. *PLOS ONE*, 15(2), e0229436. <https://doi.org/10.1371/journal.pone.0229436>
- Kim, D., Lee, D., & Jang, S. (2016). Technological novelty profile and invention's future impact. *EPJ Data Science*, 5(1), 11. <https://doi.org/10.1140/epjds/s13688-016-0069-1>
- Schilling, Melissa A. and Phelps, Corey C. and Phelps, Corey C., Small World Networks and Knowledge Creation: Implications for Multiple Levels of Analysis (January 28, 2004). <http://dx.doi.org/10.2139/ssrn.385022>
- Tóth, G., & Lengyel, B. (2021). Inter-firm inventor mobility and the role of co-inventor networks in producing high-impact innovation. *Journal of Technology Transfer*, 46, 117–137. <https://doi.org/10.1007/s10961-019-09758-5>
- Tóth, G., Juhász, S., Elekes, Z., & Lengyel, B. (2021). Repeated collaboration of inventors across European regions. *European Planning Studies*, 29(12), 2252–2272. <https://doi.org/10.1080/09654313.2021.1914555>
- Uzzi, B., Mukherjee, S., Stringer, M., & Jones, B. (2013). Atypical combinations and scientific impact. *Science*, 342(6157), 468–472. <https://doi.org/10.1126/science.1240474>
- Verhoeven, D., Bakker, J., & Veugelers, R. (2016). Measuring technological novelty with patent-based indicators. *Research Policy*, 45(3), 707–723. <https://doi.org/10.1016/j.respol.2015.11.010>